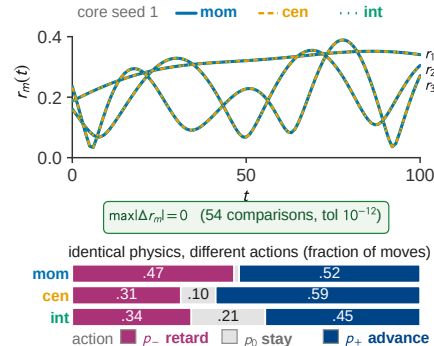
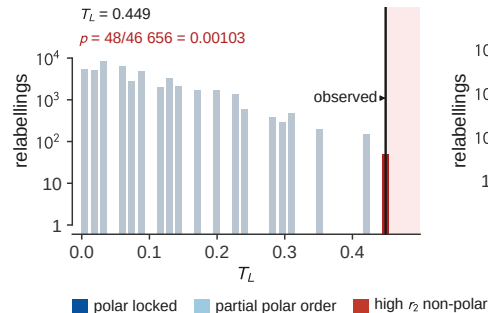


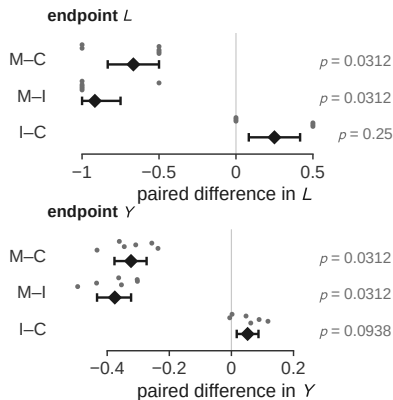
A Exact $K=0$ engine control



D Exact null for T_L

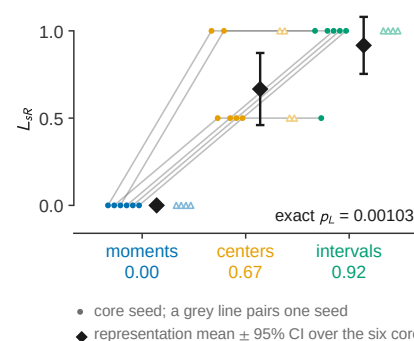


G Pairwise seed-level contrasts

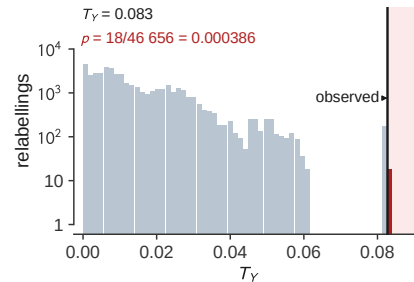


Grey dots: the six core-seed differences.
Diamond: mean with the stored bootstrap 95% CI.
 p : exact sign-flip test over all 64 sign patterns.

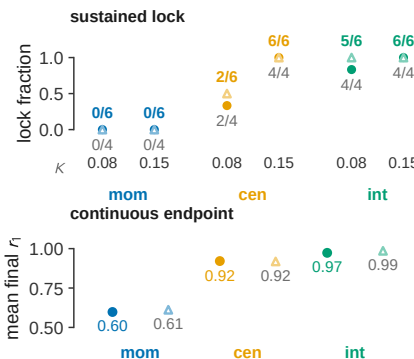
B Sustained-lock score L



E Exact null for T_Y

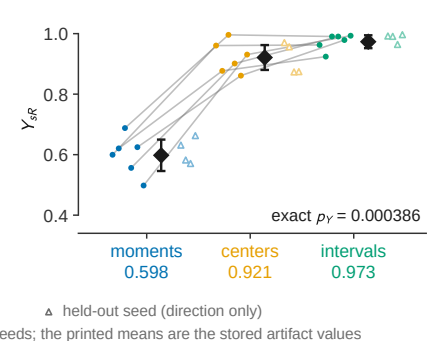


H Core vs held-out direction

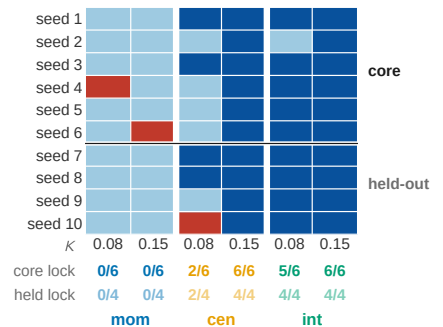


Filled circle: core six. Open triangle: held-out four.
Counts above each column are core, below are held-out.

C Final polar order γ



F Phenotype matrix



I Prespecified gate verdicts

PASS $K=0$ control

engine and matching valid
 $\max|\Delta r_m| = 0$, $\text{tol } 10^{-12}$

PASS Gate A

macro representation effect
 $p_L = 0.00103$, $p_Y = 0.000386$

PASS Gate B

qualitative phase separation
lock fraction at $K = 0.08$: 0.00, 0.33, 0.83

FAIL Gate C

GPT phenotype map not replicated
final r_1 reverses: $0.60 < 0.92 < 0.97$